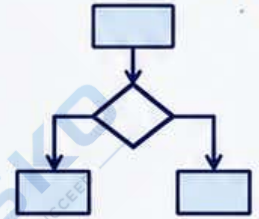
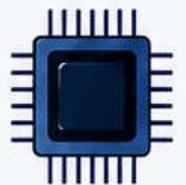


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Most Important Questions

INSTRUCTIONS



1. Understand the Concepts

Go through the basic concepts related to each question before practising.



2. Practice Regularly

Solve these important questions and their variations for better understanding.



3. Focus on Exam Pattern

These questions are selected as per previous year papers and latest syllabus.



4. Attempt Within Time Limit

Practice under timed conditions to improve speed and accuracy.



5. Revise and Self-Check

Review your answers, identify weak areas and revise regularly to score higher.

Learn • Practice • Improve • Succeed

YOUR HARD WORK PAYS OFF!



DSM — Most Important Questions

Unit I - Binary Arithmetic & Number Systems

- **Answers in one or two lines (1 mark questions)**

1. What is Binary Number System?
2. What is Octal Number System?
3. What is Hexadecimal Number System?
4. What is 1's complement?
5. What is 2's complement?
6. What is Multiplication in binary?
7. What is Division in binary?
8. What is BCD?
9. What is Excess-3 code?
10. What is Gray code?

- **Short Answer Questions (3 marks questions)**

1. Explain conversion between Binary, Octal and Hexadecimal number systems.
2. Explain binary addition and subtraction using 1's and 2's complement.
3. Explain binary multiplication and division.
4. Explain BCD, Excess-3 and Gray codes.
5. Convert numbers between different number systems with suitable examples.

- **Long Answer Questions (6 marks questions)**

1. Explain Binary, Octal and Hexadecimal number systems and their conversions with examples.
2. Solve binary addition and subtraction using 1's and 2's complement.
3. Explain binary multiplication and division with suitable examples.
4. Explain BCD, Excess-3 and Gray codes with examples

Unit II - Logic Gates

- **Answers in one or two lines (1 mark questions)**

1. What is a Logic Gate?
2. What is a NOT gate?
3. What is an AND gate?
4. What is an OR gate?
5. What is a NAND gate?
6. What is a NOR gate?
7. What is an EX-OR gate?
8. What is an EX-NOR gate?
9. What is a Universal Gate?
10. What is a Half Adder?
11. What is a Full Adder?

- **Short Answer Questions (3 marks questions)**

1. Draw and explain NOT, AND and OR gates with truth tables.
2. Draw and explain NAND and NOR gates with truth tables.
3. Explain EX-OR and EX-NOR gates with truth tables.
4. Explain NAND and NOR as Universal Gates.
5. Explain Half Adder with truth table and logic circuit.
6. Explain Full Adder with truth table and logic circuit.

- **Long Answer Questions (6 marks questions)**

1. Explain all basic logic gates with symbols and truth tables.
2. Explain Universal Gates and implement basic gates using NAND/NOR.
3. Explain Half Adder and Full Adder with truth tables and logic diagrams.
4. Design basic gates using NAND and NOR gates.

Unit III - Boolean Algebra & K-Maps

- **Answers in one or two lines (1 mark questions)**

1. What is Boolean Algebra?
2. State any two Boolean laws.
3. What is De Morgan's theorem?
4. What is a Boolean equation?
5. What is a Product Term?
6. What is a Sum Term?
7. What is a Fundamental Product?
8. What is a Fundamental Sum?
9. What is a K-Map?
10. What is a minterm?
11. What is a maxterm?

- **Short Answer Questions (3 marks questions)**

1. Explain basic Boolean Algebra laws.
2. State and prove De Morgan's theorems.
3. Explain simplification of Boolean equations using Boolean Algebra.
4. Explain Fundamental Products and Sum Terms.
5. Explain K-Map and its purpose.
6. Explain 2-variable, 3-variable and 4-variable K-Maps.

- **Long Answer Questions (6 marks questions)**

1. Explain Boolean Algebra laws and simplify Boolean expressions using them.
2. Explain and prove De Morgan's theorems with examples.
3. Simplify Boolean expressions using K-Maps for 2, 3 and 4 variables.
4. Explain Fundamental Products and Sum Terms with suitable examples.
5. Solve Boolean expressions using K-Map and obtain the simplified expression.

Unit IV - Microprocessor

- **Answers in one or two lines (1 mark questions)**

1. What is a Microprocessor?
2. What is a Microcomputer?
3. What is the block diagram of a microcomputer?
4. What is an 8085 microprocessor?
5. How many pins does the 8085 have?
6. What is the function of ALU?
7. What is the function of the Control Unit?
8. What is the function of Registers?
9. What is the address bus?
10. What is the data bus?

- **Short Answer Questions (3 marks questions)**

1. Explain the block diagram of a microcomputer.
2. Explain the architecture of the 8085 microprocessor.
3. Explain the main features of 8085.
4. Explain the pin configuration of 8085.
5. Explain the functions of important pins of 8085.

- **Long Answer Questions (6 marks questions)**

1. Draw and explain the architecture/block diagram of the 8085 microprocessor.
2. Explain the 8085 pin configuration in detail.
3. Explain the block diagram of a microcomputer and its components.
4. Explain the evaluation/features of the 8085 microprocessor.